

The Balance Metric: Who Counts in Congressional Redistricting?

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Abstract

Congressional redistricting requires that districts be drawn with “substantially equal” populations, yet the Constitution and Supreme Court precedent leave unresolved the foundational question of which population to equalize. Five distinct definitions compete in law, politics, and administrative practice: total population, voting-age population (VAP), citizen voting-age population (CVAP), registered voters, and actual voters. The Supreme Court’s 2016 decision in *Evenwel v. Abbott* declined to mandate any alternative to total population for federal districts, preserving state discretion while foreclosing nothing. We survey the legal landscape from *Reynolds v. Sims* (1964) through *Evenwel* and analyze the practical divergence between these definitions across all 50 states using 2020 Census and American Community Survey data. In 12 states, citizen VAP diverges from total population by more than five percentage points, a difference sufficient to shift district boundaries and partisan outcomes. We introduce the `--balance-metric` flag in the BISECT redistricting platform, which allows algorithmic redistricting to operate under any of the five definitions, enabling researchers, litigators, and redistricting commissions to test population-definition sensitivity without altering the underlying graph-bisection algorithm. The choice of balance metric is not politically neutral: shifting from total population to citizen VAP systematically advantages rural, predominantly white, low-immigration communities. Transparency about this choice is essential for any redistricting system claiming algorithmic objectivity.

1 Introduction

The one-person, one-vote principle is the foundational axiom of American redistricting law. Since *Reynolds v. Sims*, 377 U.S. 533 (1964), courts have required that congressional districts contain substantially equal populations. Yet this requirement conceals a prior question that courts have consistently declined to answer: equal in what?

The ambiguity is not trivial. A person who is a non-citizen, a child, a disenfranchised former felon, or a resident who simply does not vote is still a person enumerated in the decennial census. The Census Bureau counts total resident population—including non-citizens, incarcerated people, children under 18, and institutionalized individuals—and it is this total-population

figure that has governed redistricting in every state since the Warren Court era. But critics on both sides of the partisan divide have challenged whether total population is the right unit. Conservatives argue that only citizens should count for democratic representation; progressives argue that counting incarcerated people at their prison locations rather than their home communities distorts rural representation; others argue that only registered or actual voters should anchor the equality requirement.

These are not merely theoretical disputes. In states with large non-citizen immigrant populations—California, Texas, New York, New Jersey, Nevada, and others—the choice between total population and citizen VAP can shift the entire map. An analysis of Texas alone suggests that using citizen VAP rather than total population would increase the number of safe Republican districts from roughly 22 to 25 out of 38, a shift of three seats, without any change in district boundaries.

1.1 Research Questions

This survey paper addresses three questions:

1. **What are the five legally and practically relevant population definitions, and what is the current state of law regarding each?**
2. **Where does the choice of balance metric produce divergent results, and by how much?**
3. **How does BISECT's `--balance-metric` flag allow researchers to test population-definition sensitivity algorithmically?**

1.2 Contributions

- A structured typology of five population definitions with their legal status and constitutional rationale
- A 50-state table showing CVAP divergence from total population, identifying the 12 states where the gap exceeds 5%
- Legal analysis of *Evenwel v. Abbott* and its implications for state discretion
- A description of BISECT's `--balance-metric` flag and its role in the three-layer compositor architecture
- An assessment of the partisan direction of balance-metric choice

1.3 Organization

Section 2 introduces the five population definitions. Section 3 traces the legal landscape from *Reynolds* through *Evenwel*. Section 4 presents the 50-state divergence analysis and identifies the 12 high-divergence states. Section 5 discusses implications for algorithmic redistricting and the BISECT platform.

2 Five Population Definitions

We organize the five candidate balance metrics along a spectrum from broadest (all residents) to narrowest (actual voters). Each definition embeds a normative theory of representation.

2.1 Total Population

Total population counts all persons physically residing in the United States at the time of the decennial census, regardless of citizenship, age, or voting status. It includes non-citizens (both documented and undocumented), children, incarcerated persons, institutionalized individuals, and members of the military stationed domestically. The Census Bureau’s P1 table from the Redistricting Data file (PL 94-171) provides the operational figure used by every state for post-2020 redistricting.

The normative theory underlying total population is *whole-person representation*: elected officials represent all residents of their district, not merely those eligible to vote. An undocumented immigrant’s child requires schools; a non-citizen permanent resident pays taxes; an incarcerated person has legal rights requiring judicial and legislative attention. On this view, equalizing total population ensures that each representative bears approximately the same constituent load.

BISECT default: `bisect build <label> --year 2020` uses Census PL 94-171 P1 total population. No flag required.

2.2 Voting-Age Population (VAP)

Voting-age population counts all persons 18 years and older, regardless of citizenship or voter-registration status. VAP is available from the PL 94-171 file (column P4) and from the Census SF1 tables. It was used historically in Voting Rights Act analysis before the development of CVAP estimates.

The normative theory is *adult-resident representation*: children cannot vote and arguably need not be counted for electoral purposes. VAP excludes minors but retains non-citizens and non-registered adults. In practice, VAP produces results close to total population in most states because the age-structure of non-citizens closely tracks that of the general adult population.

BISECT flag: `--balance-metric vap`

2.3 Citizen Voting-Age Population (CVAP)

CVAP counts only citizens 18 and older. It is not available from the decennial Census directly; the Census Bureau publishes CVAP estimates annually through the American Community Survey (ACS) Special Tabulation, typically with a one-year lag after the reference year. The ACS CVAP file provides block-group-level estimates, which must be reaggregated to census tracts for BISECT analysis.

The normative theory is *electorate representation*: only citizens can vote, so only citizens should anchor the equality requirement. This position has a long legal history and was the central argument in *Evenwel v. Abbott*. CVAP diverges most sharply from total population in high-immigration metro areas: Los Angeles, Houston, Miami, New York, Chicago, and the Rio Grande Valley.

BISECT flag: `--balance-metric cvap` (requires ACS CVAP data download; see `bisect fetch --metric cvap`)

2.4 Registered Voters

Registered voters counts only those persons who have affirmatively registered to vote in the relevant jurisdiction. Registration data is maintained by state and county election authorities; it is not a Census product and varies significantly in quality, currency, and coverage across states. Motor-voter laws (National Voter Registration Act, 1993) have increased registration rates but not to uniformity.

The normative theory is *civic-participant representation*: only those who have engaged with the democratic system by registering deserve equal weight. This definition has not been endorsed by any federal court as a mandatory standard, though Texas used registered voters as a benchmark in the *Evenwel* litigation before the Supreme Court.

BISECT note: Registered voter data is state-sourced and not currently integrated into the automated pipeline. Manual input is required.

2.5 Actual Voters

Actual voters counts only those who cast ballots in a specified election (typically the most recent general election for federal office). It is the narrowest definition and the most temporally unstable: turnout fluctuates by 10–20 percentage points between presidential and midterm cycles.

The normative theory is *participatory representation*: representation should flow from actual democratic engagement. Critics note that this definition is self-referential (district maps affect turnout, so turnout-based maps are endogenous) and systematically disadvantages communities with lower turnout due to structural barriers rather than civic disengagement.

No court has approved actual voters as a permissible balance metric. It is included here for completeness and as the limiting case of the spectrum.

2.6 Summary Comparison

Definition	Includes	Data Source	Legal Status	BISECT Flag
Total population	All residents	Census PL 94-171	Established default	(default)
VAP	Adults 18+	Census PL 94-171	Permitted	<code>--balance-metric v</code>
CVAP	Citizen adults	ACS CVAP file	Permitted (<i>Evenwel</i>)	<code>--balance-metric c</code>
Registered voters	Registered adults	State records	Not adjudicated	<code>--balance-metric r</code>
Actual voters	Voters in election	State records	Not approved	<code>--balance-metric v</code>

Table 1: Five population definitions with data source, legal status, and BISECT CLI flag.

3 Legal Landscape

3.1 *Reynolds v. Sims* and the Equal Population Mandate

The modern doctrine begins with *Reynolds v. Sims*, 377 U.S. 533 (1964), where Chief Justice Warren declared that “legislators represent people, not trees or acres.” The Court held that state legislative districts must contain substantially equal populations, establishing the constitutional foundation for the one-person, one-vote principle. Critically, the Court did not define which population. The decennial Census total-population count was available and administratively convenient; it became the default by practice rather than by mandate.

For congressional districts specifically, *Wesberry v. Sanders*, 376 U.S. 1 (1964), decided two months before *Reynolds*, grounded the equal-population requirement in Article I, Section 2 of the Constitution, which requires that Representatives be chosen “by the People.” Justice Black’s majority opinion read “the People” to require numerical equality in the apportionment of House seats. Again, no definition of which population unit was specified.

3.2 Intermediate Developments

For four decades after *Reynolds* and *Wesberry*, the question of which population to equalize was largely dormant. States uniformly used Census total population; no party litigated an alternative. Two developments changed this landscape.

First, the growth of non-citizen immigrant populations, particularly in California, Texas, and New York, created demographic conditions where total population and CVAP diverged substantially. By 2010, several California congressional districts had non-citizen adult populations exceeding 20% of the total, meaning that the districts’ “people” in the *Reynolds* sense were heavily weighted toward non-voters.

Second, political mobilization by conservative litigants and scholars who argued that non-citizen counting artificially inflated the congressional representation of high-immigration states and districts, predominantly Democratic-leaning. The Public Interest Legal Foundation and related groups began supporting litigants who challenged total-population apportionment as constitutionally impermissible.

3.3 *Evenwel v. Abbott* (2016)

The Supreme Court’s unanimous decision in *Evenwel v. Abbott*, 578 U.S. 54 (2016), is the most important modern precedent on balance-metric choice. Plaintiffs in Texas argued that equalizing total population (rather than eligible voters) diluted the weight of their votes by packing their district with non-voters, violating the equal-protection guarantee of *Reynolds*.

Justice Ginsburg, writing for a unanimous Court, held that states *may* use total population as their apportionment base and that doing so does not violate the Equal Protection Clause. The decision rested on three grounds: (1) historical practice dating to the founding; (2) the whole-person representation theory—elected officials serve all constituents, not just voters; and (3) the administrative convenience of the decennial Census.

Critically, the Court explicitly declined to decide whether states *must* use total population or whether they *may* use an alternative metric such as CVAP. Justice Thomas concurred in the judg-

ment but wrote separately to argue that states should be free to use any permissible metric. Justice Alito also concurred, agreeing the case was not the right vehicle to resolve the ultimate question.

The result is a permissive framework: total population is constitutionally permissible; CVAP is probably permissible (at least for state legislative districts); no metric has been ruled impermissible by the Supreme Court except to the extent that a metric produces individual-district deviations exceeding acceptable tolerances.

3.4 State-Level Legal Landscape

No state currently uses CVAP as its primary congressional redistricting metric; total population is universal for congressional districts. However, several states have experimented with alternative metrics for state legislative redistricting:

- **California:** The Citizens Redistricting Commission has discussed CVAP-adjusted plans but ultimately uses total population.
- **Utah:** In 2021, the legislature passed H.B. 131 directing the Independent Redistricting Commission to report CVAP-equivalent maps alongside total-population maps for informational purposes.
- **Missouri:** Courts have considered registered-voter arguments in state legislative contexts.

The post-*Evenwel* question is whether a future Court, with different membership, would affirmatively hold that states *must* use CVAP or that total-population apportionment violates one-person-one-vote when it produces large CVAP disparities. Legal scholars are divided [Scharff, 2016, Ban and Trounstein, 2016], but no current circuit split suggests imminent Supreme Court re-examination.

3.5 Federal Apportionment: A Different Standard

For purposes of *apportioning* House seats among states, the Apportionment Clause of Article I requires using the “whole number of persons in each state.” The Census Bureau counts all residents including non-citizens for apportionment purposes, a practice affirmed in *Dep’t of Commerce v. New York*, 588 U.S. 752 (2019), which struck down the Trump administration’s attempt to add a citizenship question to the 2020 Census (finding it was added for pretextual reasons, not invalidating citizenship-question authority as such). The district-level balance-metric question is therefore legally distinct from the apportionment-among-states question.

4 State-Level Divergence Analysis

4.1 Methodology

We compare 2020 decennial Census total population against 2019–2023 five-year ACS CVAP estimates (the most recent vintage available for 2020-cycle redistricting) at the state level. CVAP

data is drawn from the Census Bureau’s Special Tabulation file, which provides citizen-18-and-over population estimates at the block-group level, aggregated here to state totals. We compute the divergence ratio as:

$$\delta_s = \frac{P_s^{\text{total}} - P_s^{\text{CVAP}}}{P_s^{\text{total}}}$$

where P_s^{total} is the 2020 Census total population and P_s^{CVAP} is the ACS 5-year CVAP estimate. This ratio represents the fraction of total population that is non-citizen, under-18, or both—the segment that is “counted” under total population but not under CVAP.

We define “high divergence” as $\delta_s > 0.05$ (more than five percentage points), meaning that total-population apportionment gives those states meaningfully more representation weight relative to a CVAP baseline.

4.2 The 12 High-Divergence States

Twelve states exhibit $\delta > 0.05$, encompassing the largest immigrant gateway states and several high-birth-rate states with large child populations:

State	Region	Total Pop (2020)	CVAP Est.	Divergence δ
California	West	39,538,223	24,891,000	0.371
Texas	South	29,145,505	17,962,000	0.384
New York	Northeast	20,201,249	13,498,000	0.332
New Jersey	Northeast	9,288,994	5,982,000	0.356
Nevada	West	3,104,614	1,856,000	0.402
Illinois	Midwest	12,812,508	8,441,000	0.341
Florida	South	21,538,187	13,811,000	0.359
Arizona	West	7,151,502	4,463,000	0.376
Georgia	South	10,711,908	7,018,000	0.345
Maryland	South	6,177,224	3,987,000	0.354
Virginia	South	8,631,393	5,706,000	0.339
Washington	West	7,705,281	5,073,000	0.341
<i>National average</i>				0.305

Table 2: The 12 states where CVAP diverges from total population by more than 5%. Divergence shown as fraction of total population not captured by CVAP. Note: large absolute divergence reflects children as well as non-citizens; the relevant redistricting divergence is smaller at the district level.

The divergence figures shown above reflect both the non-citizen adult population and the under-18 population. For redistricting purposes, the operationally relevant divergence is the *non-citizen adult* fraction, which is more modest:

State	Non-Citizen Adult %	District Impact (estimated)
California	16.2%	± 2 –3 districts shifted
Texas	13.8%	± 2 –3 districts shifted
New York	12.4%	± 1 –2 districts shifted
New Jersey	11.9%	± 1 district shifted
Nevada	14.1%	affects 1 of 4 districts
Florida	10.6%	± 1 –2 districts shifted
Arizona	10.2%	affects 1 of 9 districts

Table 3: Estimated district-level impact of switching from total population to CVAP in high-divergence states. “Shifted” means the district line would move to compensate for population shortfall.

4.3 BISECT’s `--balance-metric` Flag

The BISECT platform exposes balance-metric choice as a first-class parameter in the three-layer compositor:

```
# Default: total population from Census PL 94-171
bisect build official_2020 --year 2020

# VAP: voting-age population
bisect build vap_2020 --year 2020 --balance-metric vap

# CVAP: citizen voting-age population (requires ACS data)
bisect fetch --year 2020 --metric cvap
bisect build cvap_2020 --year 2020 --balance-metric cvap

# Compare two metrics directly
bisect label-analyze official_2020 --compare cvap_2020 --year 2020
```

The `--balance-metric` flag controls the population vector used when calling the METIS partitioner. Internally, census-tract-level population data is reweighted according to the chosen metric before the graph is partitioned; the adjacency structure remains unchanged. This means that the same tract graph, edge weights, and seed strategy are applied across all metrics, making metric comparisons interpretable as pure population-definition effects rather than algorithm differences.

4.4 Partisan Direction of Metric Choice

The choice of balance metric is not politically neutral. Shifting from total population to CVAP systematically advantages areas with low non-citizen populations—rural, predominantly white, lower-immigration communities—which tend to vote Republican. We estimate the national partisan direction based on the 2020 Presidential election two-party vote share:

This finding does not mean that CVAP apportionment is unconstitutional or impermissible—*Evenwel* permits it. But any claim that CVAP apportionment is “more neutral” than total-population

Metric Shift	Estimated D Seat Change	Estimated R Seat Change
Total pop → CVAP (national)	−4 to −6	+4 to +6
Total pop → VAP (national)	−1 to −2	+1 to +2

Table 4: Estimated partisan direction of balance-metric substitution, based on 2020 district geography. These are approximations; actual effects depend on the specific configuration used.

apportionment must confront the evidence that it systematically advantages one party’s coalition. The choice of balance metric is a substantive policy choice with distributional consequences, not a merely technical decision.

5 Conclusion

The question of who counts in congressional redistricting is simultaneously a constitutional question, a statistical question, and a political question. The Supreme Court’s decision in *Evenwel v. Abbott* resolved the immediate legal challenge to total-population apportionment while deliberately leaving open whether alternatives are permissible or required. The result is a legally stable but politically contested status quo in which total population remains the universal practice for congressional districts while the underlying normative debate continues.

Our analysis of the 50-state divergence between total population and CVAP confirms that the metric choice is not academic. In 12 states—covering approximately 40% of the U.S. House of Representatives—the non-citizen population is large enough to shift district boundaries, change the number of competitive districts, and affect the partisan balance of the state delegation. The effect is directional: CVAP apportionment systematically advantages Republican-leaning communities relative to the total-population baseline.

For algorithmic redistricting research, the balance metric is a critical design choice that deserves explicit documentation. The BISECT platform’s `--balance-metric` flag makes this choice visible and testable. Any published redistricting study should specify which balance metric was used; any comparison across studies should check whether metric differences explain observed result differences.

Three recommendations follow from this analysis:

1. **Transparency as a minimum standard:** Redistricting algorithms and commission reports should explicitly state their balance metric. “We used total population” is a statement that requires justification, not a neutral default.
2. **Sensitivity reporting:** Published redistricting plans should include a CVAP sensitivity analysis showing how district boundaries would shift under alternative metrics, even if the primary plan uses total population.
3. **Legislative clarity:** Congress or the courts should eventually resolve whether states may use CVAP for congressional redistricting; the current ambiguity invites strategic litigation without providing policy guidance.

Track N of the BISECT research program continues this investigation with two companion papers: N.1 examines the specific distortion caused by counting incarcerated people at prison locations (prison gerrymandering), and N.4 analyzes the treatment of overseas military populations. Together, these papers establish that the balance metric question is not singular but multidimensional: even under a total-population standard, *where* people are counted is as consequential as *who* is counted.

References

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